

Thank you for purchasing a FANUC Robot.
READ THIS FIRST
Use this guide to unpack, install, turn on, and jog your new FANUC robot.

The appropriate level of safety for your application and installation can best be determined by safety system professionals. A qualified electrician should apply power. FANUC recommends that each customer consult with such professionals in order to provide a safe application, and take the necessary FANUC training course to operate the robot safely. Refer to your FANUC documentation provided with your robot for safety procedures.

Required Tools

- ◆ Straight-head screwdriver ◆ Torque wrench
- ◆ Cross-head screwdriver ◆ Metric hex-head wrenches

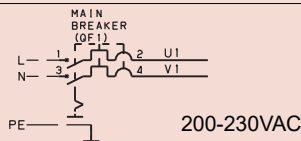
Required Items

- ◆ Properly rated transport equipment as specified
- ◆ Customer-supplied properly rated power source as specified on the Power Requirements Label
- ◆ Multimeter
- ◆ Loctite #242
- ◆ Rigid platform to hold robot
- ◆ External e-stop switch
- ◆ Safety fence/gate switch
- ◆ Customer-supplied Ethernet cable
- ◆ Laptop or PC with IE, Chrome, or Safari browser
- ◆ 4-M8 bolts (SR-3iA)
- ◆ 4-M8 flat washers (SR-3iA)
- ◆ 4-M10 bolts (SR-6iA)
- ◆ 4-M10 flat washers (SR-6iA)
- ◆ FANUC-supplied USB drive

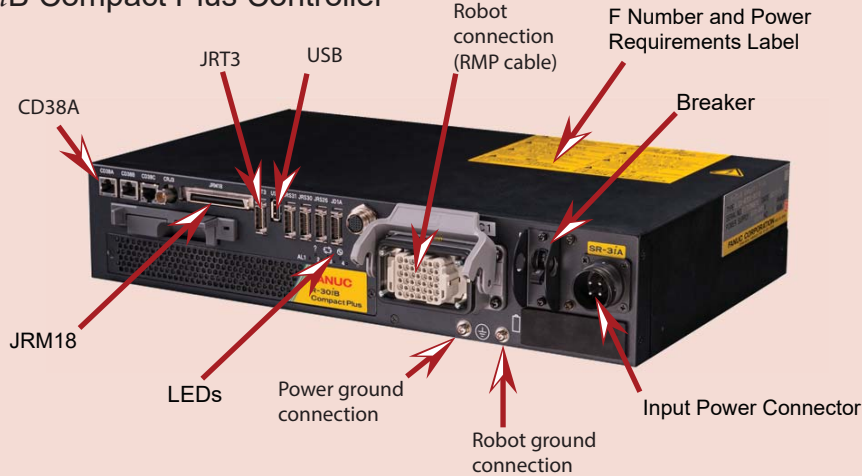
Optional Items

- ◆ Robot locator pins
- ◆ Auto Confirmation Signal switch

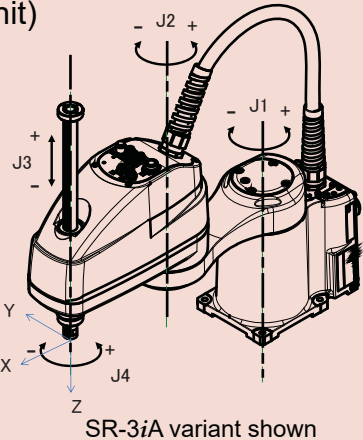
Single Phase



R-30iB Compact Plus Controller



Robot (mechanical unit)



Note: all axes are 0° in this position.

SR-3iA variant shown



A. Unpack and Install your Robot

1. Cut the yellow shipping straps, then squeeze and pull the white tabs so the outer shipping box can be removed.



2. Unbox the pallet, cut any shipping straps securing the robot, and unwrap the robot.

3. Open the large box on the pallet. In the large box you will find an accessory box, your controller and, possibly, an optional Teach Pendant and Switch Box.



4. In the accessory box you will find a hardware bag (containing fuses, jumper plugs, and labels), an Inspection Data Sheet, ground cable, and a power cable.



Store the Inspection Data Sheet in a safe, known location.

5. Remove the bolts securing the robot to the transportation equipment (shown below).

6. Use properly rated equipment to move the robot to its new location. The robot alone can weigh as much as 30kg.

WARNING

Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.

Crane capacity: 100kg or more
Sling capacity: 100kg or more

Transport posture
J1: -20°
J2: 145°
J3: -100 mm
J4: 0°

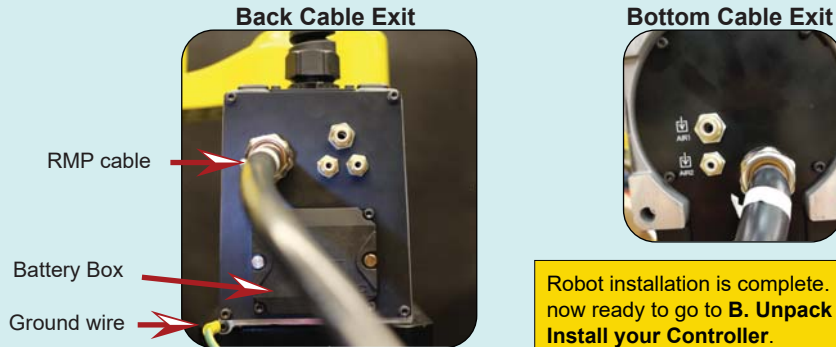
7. Fasten the robot securely to the base plate (tightening torque: SR-3iA, 23-32 Nm; SR-6iA, 46-66 Nm). Use locator pins to secure the robot precisely if it is anticipated that repeated installations will be necessary in the future for service/maintenance.

8. Remove the transportation hardware (plate for transportation, transportation equipment, eye bolts, and other hardware). Store these items and the robot skid in a safe, known location. These items are needed for future transportation.



Plate for Transportation

9. Connect the ground wire to the location on the robot base shown below.
Note: The RMP cable is attached to the robot at the factory.



Robot installation is complete. You are now ready to go to **B. Unpack and Install your Controller**.

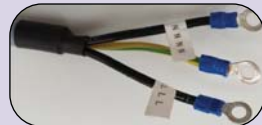
B. Unpack and Install your Controller

10. If you have not already done so, remove the controller from the box and place it in a safe location outside of the robot work area such that the grounding wire and robot connection cable can reach the robot.

11. Attach the RMP cable from the robot to the controller. (Refer to controller picture to the left.)

12. Attach the other end of the ground wire that you attached to the robot in Step 9 to the controller.

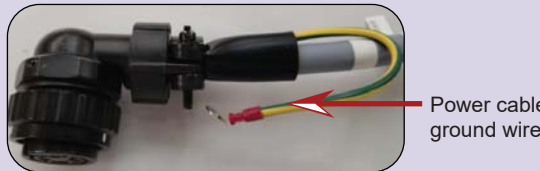
13. Have a qualified electrician connect the FANUC-supplied power cable (wires marked L, N, and ground) to an appropriate external power source.



WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

14. Connect the other end of the FANUC-supplied power cable to the Input Power Connection, located on the right-hand side of the front of the controller. Don't forget to connect the power cable ground wire (attached to the Robot Connection cable) to the power ground connection on the controller.

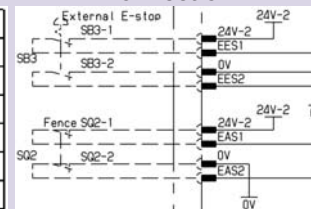


15. Wire an E-stop switch into connector JRM18 on the controller. A partial pinout and wiring diagram showing the e-stop connections is shown on the right. For more detail see the R-30iB Compact Plus CONTROLLER MAINTENANCE MANUAL.

JRM18 Partial Pinout

	24VEXT_IN	35	0VEXT
1	24V-2	36	0V
2	24V-2	37	0V
3	24V-2	38	0V
4	24V-2	39	0V
5	SDICOM1	40	SDICOM2
6	EES1	41	EES2
7	EAS1		EAS2

External E-stop/Fence Connection



Note: If JRM18 is not being used, make sure the FANUC-supplied dummy plug is inserted in JRM18.

Note: If there is no Teach Pendant (TP), make sure the FANUC-supplied dummy plug is placed into the connector labeled TP.

Note: If the TP option is used without a switch box, add an "Auto Confirmation Signal" switch outside of the safety fence, wiring it to one of the available digital input signals on the JRM18 connector.

Note: If there is no switch box, make sure the FANUC-supplied dummy plug is inserted into the connector labeled JRT3.

16. Connect the customer-supplied Ethernet cable between the PC and connector CD38A on the controller.

Controller installation is complete. **DO NOT TURN ON THE CONTROLLER UNTIL YOU ARE INSTRUCTED TO DO SO.** Go to **C. Turn on your Controller and Update Software.**

C. Turn on your Controller and Update Software

Note: If you will be using a TP to install software, please refer to the FANUC Version 8.X, 9.X Series Software Installation Guide. If you do not have this Guide, please contact FANUC.

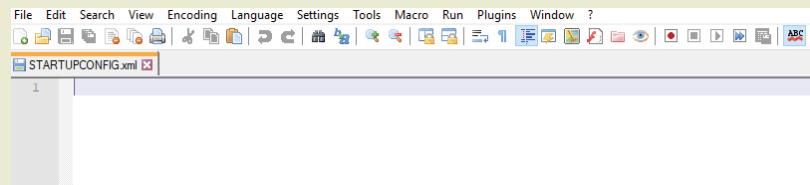
Note: If there is no need to load software (e.g. the software is the most current version available, having been loaded by FANUC America), you can skip to **G. Jog your Robot**. Otherwise, continue with step C17.

17. To ensure that the latest software is loaded, perform the following steps.

- a. Insert the FANUC-supplied USB, loaded with the robot software, into a laptop/PC. **Note:** If your PC is having problems accessing the USB, make sure your USB driver is up to date.



- b. Using a text editor (e.g. Notepad++) on the laptop/PC, create a blank XML file called STARTUPCONFIG.XML and **save it to the root** of the USB.



- c. Plug the USB into the USB port of the controller with power off.



- d. Turn the Breaker ON. When the controller power-up is in progress, the Power LED will be ON and the Cycle Start LED will blink. When the Cycle Start LED stops blinking, the start up is completed.

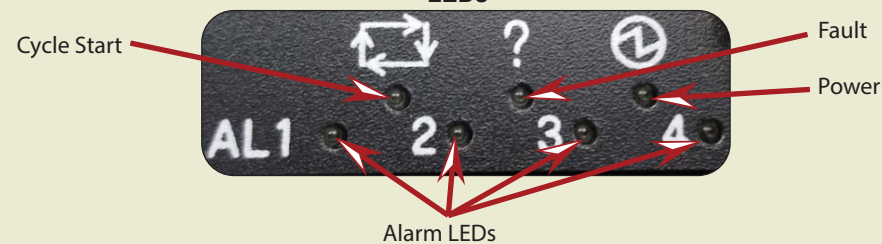
WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

Breaker

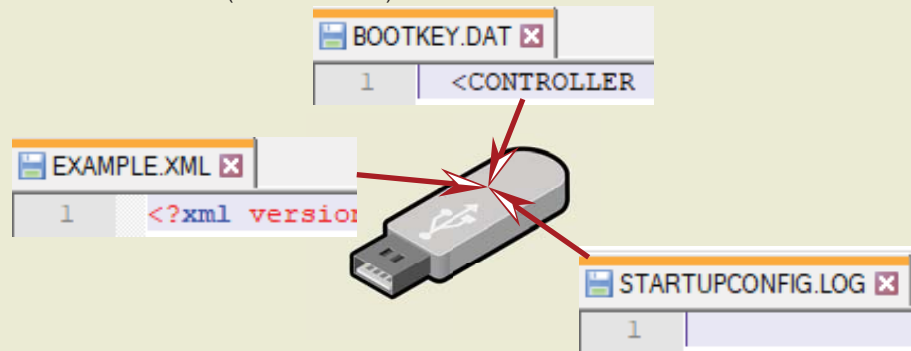


LEDs



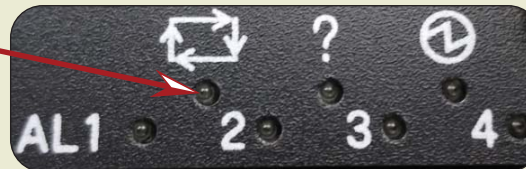
Note: Upon power-up, BOOTKEY.DAT, EXAMPLE.XML, and STARTUPCONFIG.LOG are generated in the USB (UD1: device).

- In BOOTKEY.DAT, the key that allows execution of STARTUPCONFIG.XML and CONTROLLER tags is generated. The expiration date of the key is **one hour** from its creation.
- In EXAMPLE.XML, a description example of STARTUPCONFIG.XML is generated.
- In STARTUPCONFIG.LOG, the log used during execution of the XML is written. Error information (if execution fails) is also written here.



- e. When the Cycle Start LED stops blinking, indicating that the power-up has finished, you may turn the controller off, remove the USB from the controller, and plug it into the laptop/PC. (Do not remove the USB from the controller during file writing).

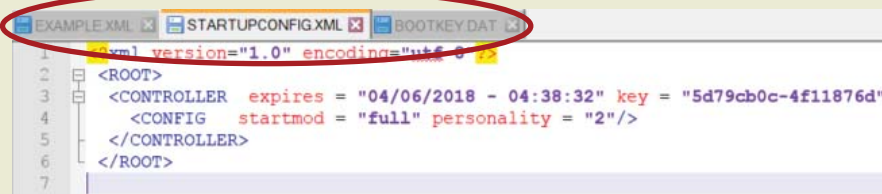
Cycle Start



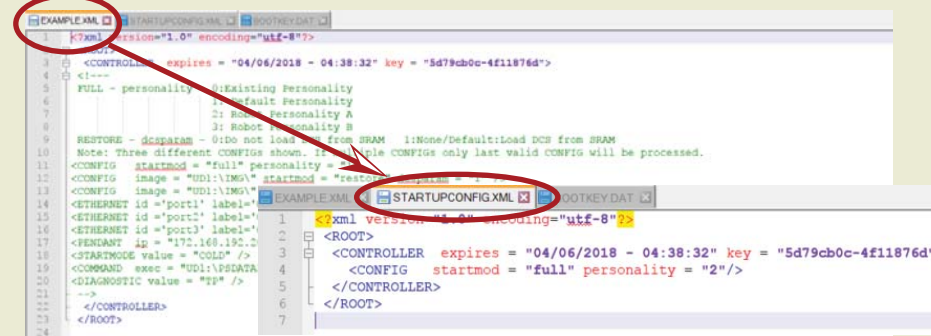
- f. Open the USB from the laptop/PC. You will see the three files named in step 17d and another one; STARTUPCONFIG.XML (see illustration of STARTUPCONFIG.XML in step 17j).

Name	Date modified	Type	Size
BOOTKEY.DAT	1/17/2018 4:05 PM	DAT File	1 KB
EXAMPLE	3/5/2018 4:56 PM	XML Document	3 KB
STARTUPCONFIG	1/17/2018 4:05 PM	Text Document	1 KB
STARTUPCONFIG	1/30/2018 2:50 PM	XML Document	1 KB

- g. Using a text editor, open EXAMPLE.XML, STARTUPCONFIG.XML, and BOOTKEY.DAT.



- h. Copy the contents of EXAMPLE.XML and paste them into STARTUPCONFIG.XML.



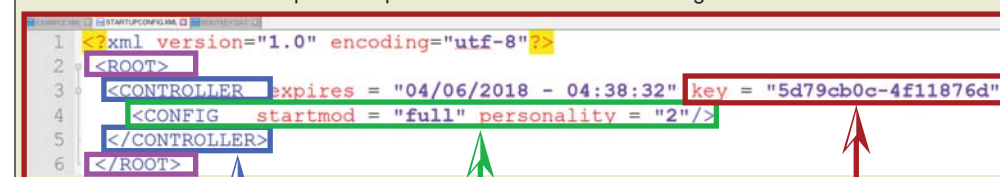
- i. Check the key in STARTUPCONFIG.XML. If it is the same as the key located in BOOTKEY.DAT, skip to the next step. If not, perform the following:
- Copy the key from BOOTKEY.DAT (e.g. key = "5d79cb0c-4f11876d").
 - Delete the key in STARTUPCONFIG.XML.
 - Paste the new key from BOOTKEY.DAT into STARTUPCONFIG.XML.
- j. Copy the instruction **<CONFIG startmod = "full" personality = "2"/>** from the commented-out instruction list in EXAMPLE.XML, and paste it after the valid key instruction in STARTUPCONFIG.XML.

Note: In the instruction **<CONFIG startmod = "full" personality = "2"/>**, the personality number that will be loaded is 2.

You can choose a different personality, simply by changing the existing personality number to the desired personality number listed in EXAMPLE.XML.

STARTUPCONFIG.XML

The controller will perform operations based on instruction tags defined in this file



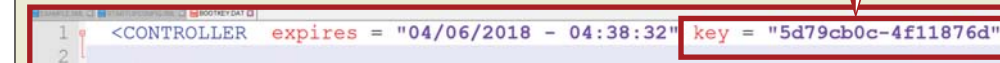
Tag Instruction for Full Software Load
The controller will execute a full software load with the following tag

Controller Element Tag
Controller instruction tags must be added between <controller> & </controller>

Root Element Tag
All tags should be added between <Root> & </Root>

Key Tag
Input a valid key in startupconfig.xml from BOOTKEY.dat
The key will expire 1 hour after generation

BOOTKEY.DAT



- k. Save STARTUPCONFIG.XML.

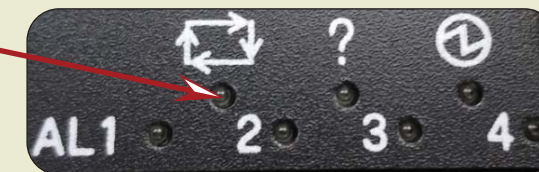
- l. Close all three files and remove the USB from the laptop/PC.

- m. Insert the USB back into the powered off controller.

- n. Turn on the controller. The XML file will be executed (software will be loaded).

- o. The software load might take in the range of 30-45 minutes; be patient. The load is finished when the Cycle Start LED on the controller stops blinking and turns off.

Cycle Start



- p. After the load has finished, remove the USB from the controller and open it on the laptop/PC.

- q. If the XML command was successfully executed, STARTUPCONFIG.XML will be renamed to OLDSTARTUPCONFIG.XML. Check to make sure that the file name was changed.

Name	Date modified	Type	Size
OLDSTARTUPCONFIG	3/26/2018 11:05 AM	XML Document	2 KB
STARTUPCONFIG	3/25/2018 10:03 PM	Text Document	1 KB
EXAMPLE	3/25/2018 9:27 PM	XML Document	2 KB

- r. Cycle power by turning the Breaker OFF and back ON.

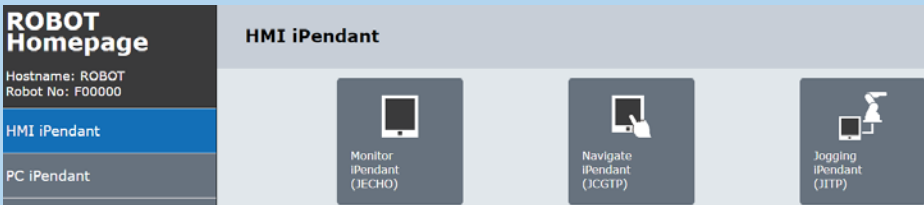
You have turned on your controller and updated software.
You are now ready to go to **D. Initiate Communication**.

Go to
Page 3

D. Initiate Communication

18. Connect an Ethernet cable between your laptop/PC and the robot.
19. Set the default address of the laptop/PC as a static IP. The PC's address should be chosen to be on the same subnet as the robot (the robot's address is 192.168.1.100). Contact your systems administrator for assistance.
20. Use a web browser (Internet Explorer (version 11 or greater), Chrome (latest version), or Safari) to connect to the robot's default IP address, 192.168.1.100. The ROBOT Homepage shown below should appear. If the screen does not appear, return to the previous step and verify that the laptop/PC and robot are on the same subnet.

Note: Initially, the robot will power-up in **Controlled Start** mode.



Note: When using IE or Chrome, make sure to clear browser cache or screens might not display properly.

IE - Press the shortcut CTRL-R

Chrome (shown on the right) - Menu (the icon with 3 periods horizontally) >> More tools >> Clear browsing data. Press CLEAR DATA.

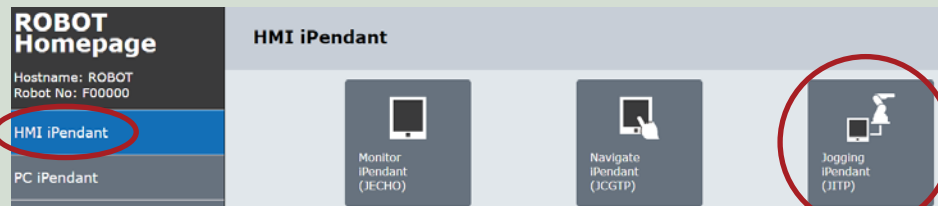


Note: If the laptop/PC will be communicating with multiple robots over a network, each robot must have a unique IP address.

The computer and robot are communicating.
You are now ready to go to **E. Master and Calibrate your Robot.**

E. Master and Calibrate your Robot

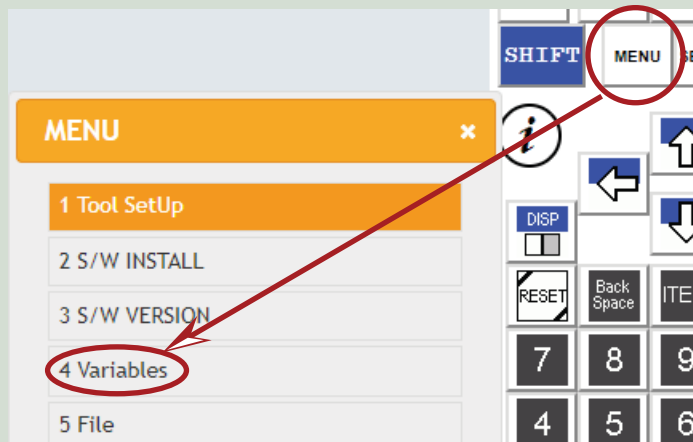
21. Make sure **HMI iPendant** is selected on the left side of the ROBOT Homepage.
22. Press **Jogging iPendant (JITP)** to open the robot Teach Pendant (TP).



23. Press **Show Keys** (located in the lower right of the TP screen) to show the TP keys.



24. Press **MENU**, and choose **Variables** from the list.



25. Cursor down to the \$DMR_GRP variable. \$DMR_GRP is typically variable 175, or close to it.

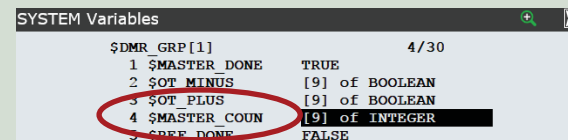
Note: On the TP keypad, you can press the down arrow key to advance through the list until you reach \$DMR_GRP (shown below). To display the variable more quickly, press **SHIFT** and then the **down arrow** to advance the cursor one page at a time.



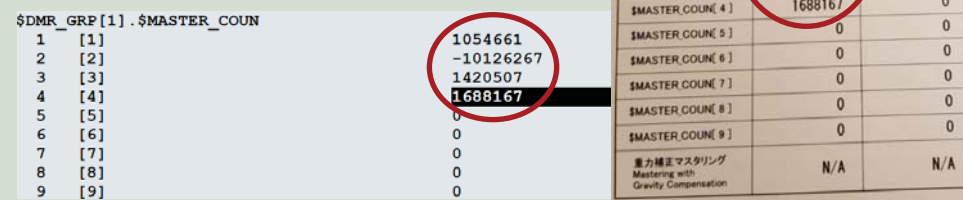
26. With your cursor on \$DMR_GRP, press **ENTER** twice.

Note: The cursor might not highlight your choice. Look at the variable number at the top of the screen to ensure you are on the correct variable (e.g. 175/834).

27. Cursor down to the \$MASTER_COUN variable, and press **ENTER**.

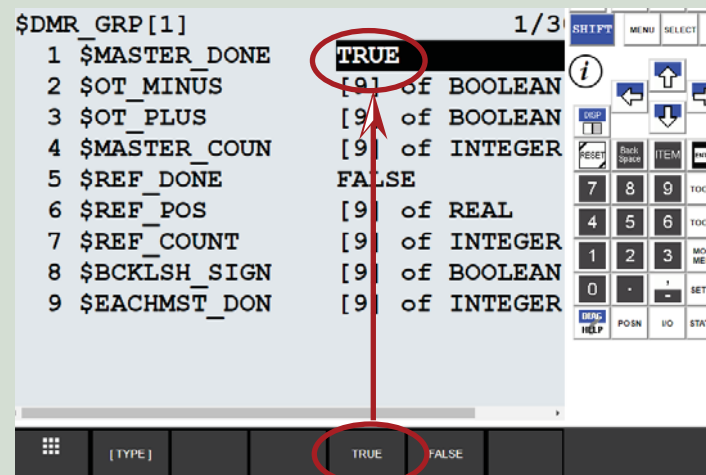


28. Using the Inspection Data Sheet from Step 4, enter each of the four Master Count numbers.

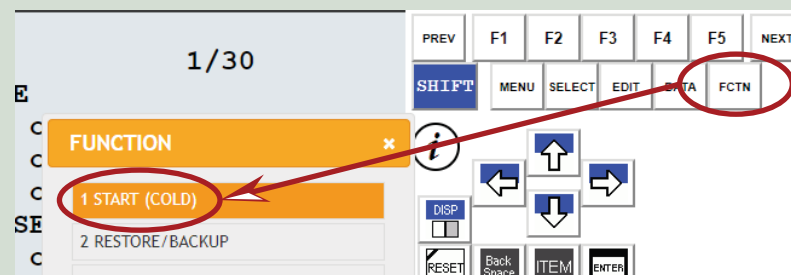


29. Press **PREV** to return to the \$DMR_GRP [1] variable list.

30. Cursor to \$MASTER_DONE, and press **TRUE** to make the variable **TRUE**.



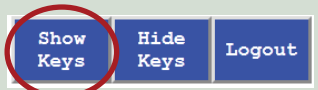
31. Press **FCTN** and choose **START (COLD)**.



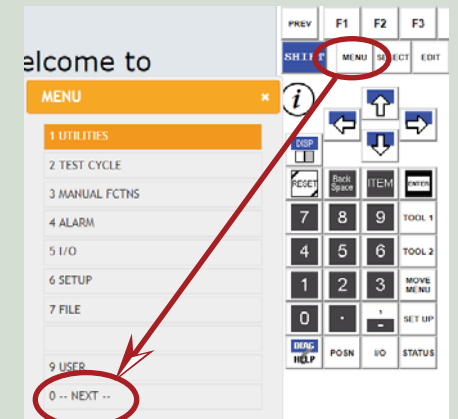
32. Using a web browser, connect to the robot's default IP address, 192.168.1.100.

33. When the homescreen appears, you will be in Cold Start mode instead of Controlled Start mode. Press **Jogging iPendant (JITP)** to display the TP.

34. Press **Show Keys** (located in the lower right of the TP screen) to show the TP keys.

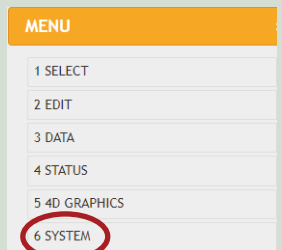


35. Press **MENU** and choose **NEXT**.



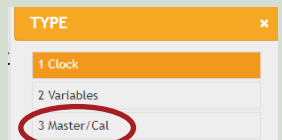
36. Select **SYSTEM** from the MENU.

Note: If you cursor to **SYSTEM**, you will have to press **ENTER** to select it.

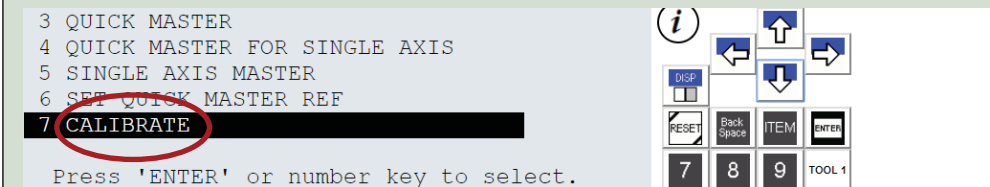


37. Press **[TYPE]** and select **Master/Cal** from the MENU.

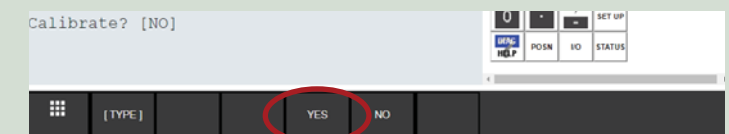
Note: If Master/Cal does not show up, choose **Variable** from the list instead. Once in the variable list, cursor to the **\$MASTER_ENB** variable, and change it from 0 to 1. Afterwards, pressing **[TYPE]** once again should enable you to see, and select, Master/Cal.



38. Cursor to and select **CALIBRATE** from the SYSTEM Master/Cal menu.



39. Press **YES** to calibrate the robot.



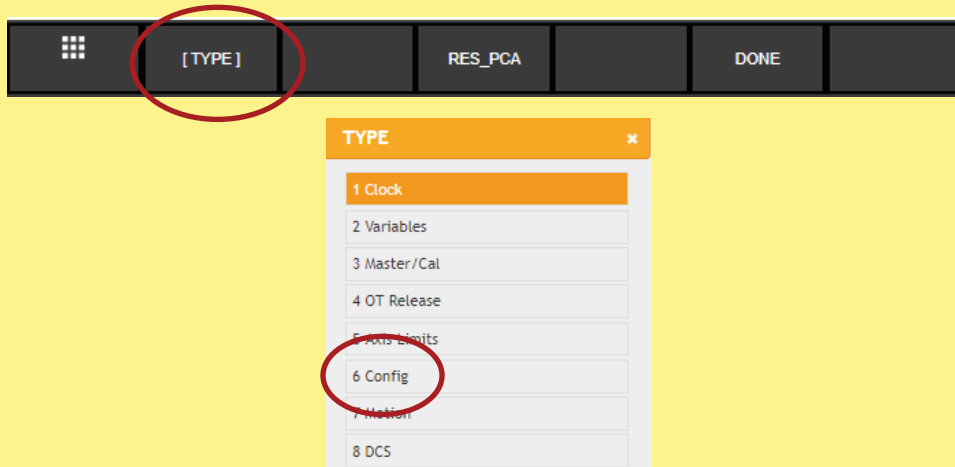
40. You should see the current position of the robot reported in joint angles as shown to the right. Check to make sure that the position displayed agrees with the actual robot position.

You have mastered and calibrated your robot. You are now ready to go to **F. Disable UI and Apply DCS Params.**

Go to Page 4

F. Disable UI and Apply DCS Params

41. Press [TYPE], and choose Config.

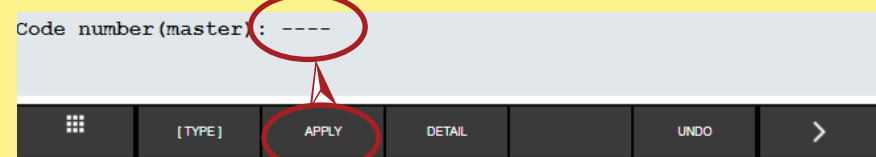


42. Cursor to **Enable UI signals** and make it **FALSE**.

7 Enable UI signals: FALSE

43. Press [TYPE] again and choose **DCS**.

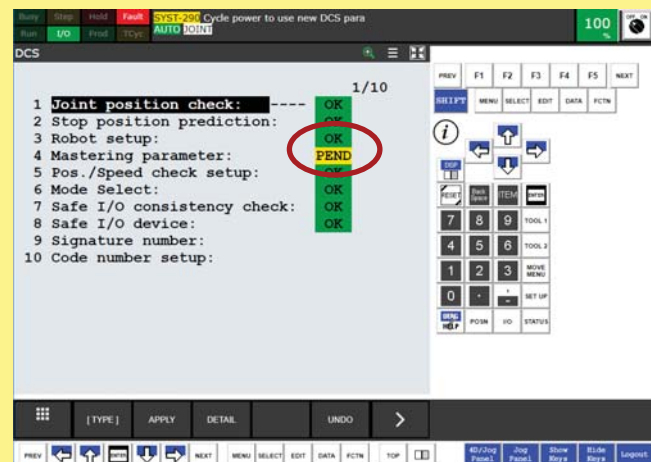
44. The next screen will indicate that there has been a change that affects DCS. Press **APPLY** to apply the Mastering change to DCS, and enter the Code number (default is 1111).



45. Press **OK**.



46. The Mastering parameter should now indicate **PEND**. Cycle power to apply the DCS change.



You have applied DCS changes. You are now ready to go to **G. Jog your Robot**.

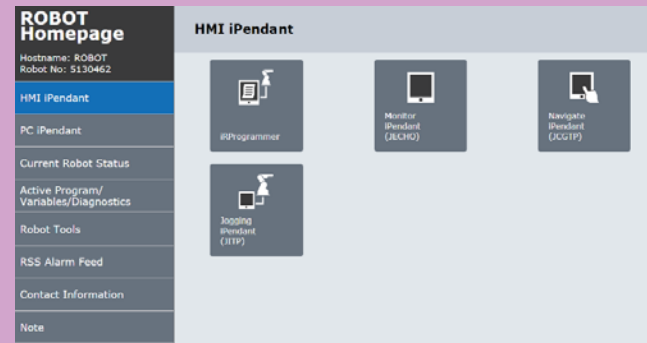
G. Jog your Robot

WARNING

When you use the jog panel to execute a jog operation, observe the following:

- The jog operation must be executed from outside the safeguards.
- Make sure that no one is inside the safety fence.
- An operational Emergency Stop button must be near the operator.
- Jogging must be executed in the sight of the robot.

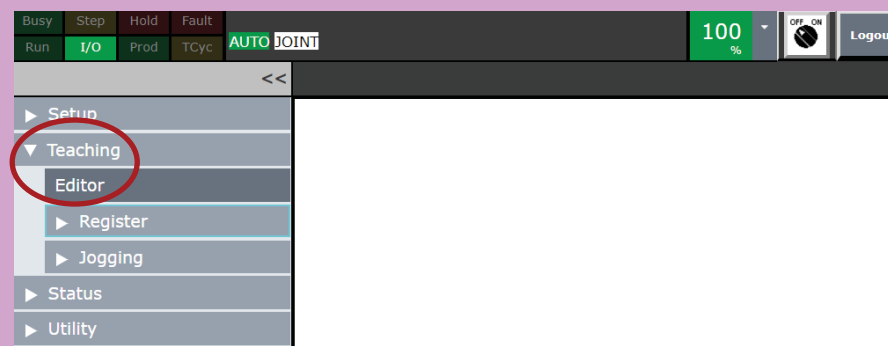
47. Use a web browser (IE, Chrome, or Safari) to connect to the robot's default IP address, 192.168.1.100. The ROBOT Homepage shown below should appear.



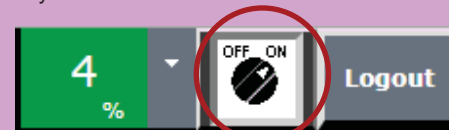
48. Make sure **HMI iPendant** is selected on the left.

49. From the ROBOT Homepage, select **iRProgrammer**.

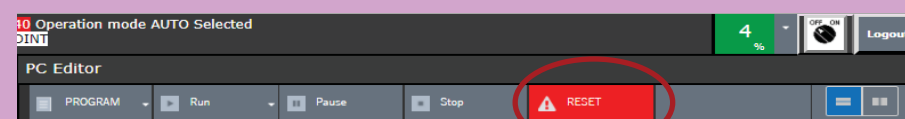
50. Select **Teaching**, and choose **Editor**.



51. Turn the TP enable key to **ON**.



52. You might have to reset a fault. If so, press the **RESET** button at the top of the screen.



53. The lower right side of the iRProgrammer PC Editor page (see **Reference** in this guide) has a jog panel that can be used to jog the robot. To move each joint axis of the robot, press each joint jog key (labeled J1 - J4) one at a time and watch the robot joint move. Refer to the Robot (mechanical unit) with joints indicated diagram on Page 1 of this guide.

54. If you wish to adjust the jogging speed, press the arrow next to the current speed (in this case 4%), and use the slider bar to adjust the speed up or down.



Note: In addition to jogging the robot, you can also move the robot by hand-dragging each axis.

CAUTION: If it is desired to move axis 4 by hand, the brake release switch must first be pressed. Be aware that, with this button pressed, gravity may cause axis 4 to move unexpectedly.

CAUTION: If a person is inside the safety fence to move the robot by hand, the power to the servo motors must be cut-off and locked out to prevent automatic operation.

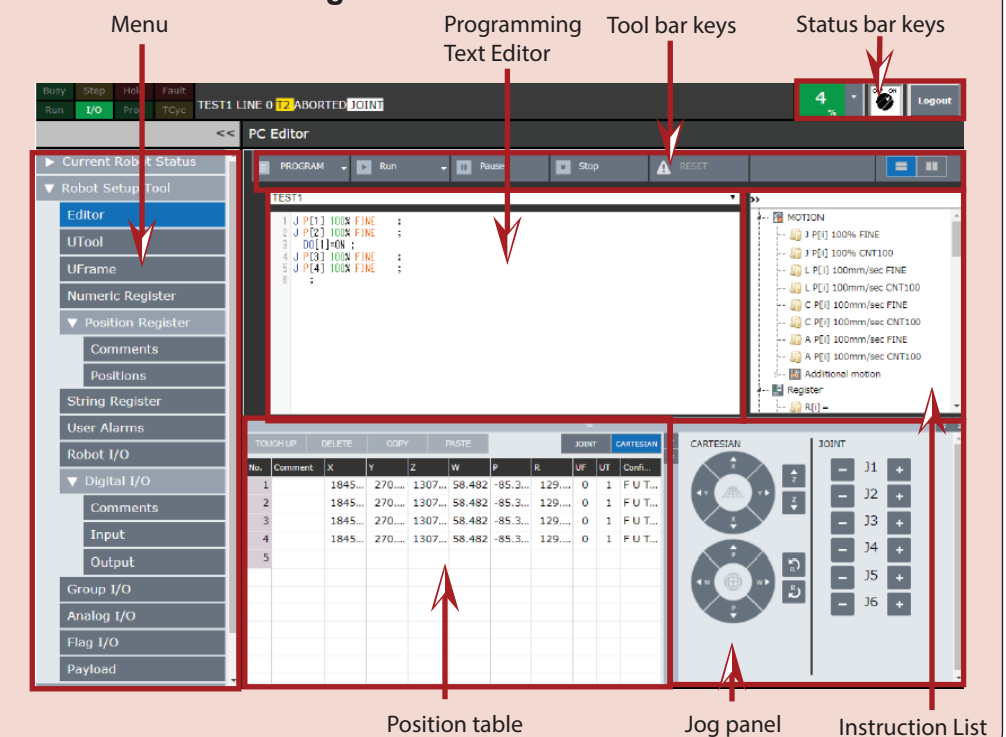
55. To hand-drag the robot, make sure the TP enable key is off and the **fault has not been reset**. Then physically grab and move the robot.



You have installed your robot. You are now ready to add tooling and program your application.

Reference

iRProgrammer™ PC Editor screen



For more information

Robot Communication
Controller Hardware/Electrical
Software Installation
Robot Hardware Operations
Software Operations
Alarm Recovery

Internet Options Setup and Operations Manual
Controller-specific Maintenance Manual
Controller-specific Software Installation Manual
Robot model-specific Operator's Manual
Application-Specific Setup and Operations Manual
Controller-specific Error Code Manual

Down Robot?

1-888-FANUC-US
Canada & United States: 1-888-326-8287

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